# Cell Site & Transceiver Data Pre-processing & Upload Guidelines (SOP)

> **Multi-Row Header Structure:** The Cell Site template utilizes a specialized **3-row composite header** (Row 1: Generation, Row 2: Technology, Row 3: Band / Property). **Data rows begin strictly on Row 4**. Never delete, merge, reorder, or alter the header rows.

## 1. Summary Checklist for Uploaders

Before uploading any Cell Site file, verify that:

- Template maintains the exact 3-row header hierarchy; data rows begin on **Row 4**.
- Required location and provider fields ( `Provider` , `Municipality` , `Province` , `Region` , `Lat` , `Long` ) are fully populated.
- **Active Technology Rule:** At least ONE band column is set to `TRUE` (or contains valid center frequency `CF: <val>, BW: <val>` ). Rows where all band columns are `FALSE` or empty cannot be used for coverage analysis.
- `Provider` matches one of the registered **Wireless Service Providers** (e.g., `DITO TELECOMMUNITY CORP.` , `GLOBE COMMUNICATIONS INC.` , `SMART COMMUNICATIONS INC.` ).
- Coordinates ( `Lat` , `Long` ) contain clean decimal numbers in WGS84 format without degree symbols or text letters.
- `Tower Type` and `AntennaType` use exact approved dropdown values.
- `Antenna Height (m)` is a pure positive numeric value (meters, no `"m"` unit).
- `Azimuth (°)` is a valid heading between `0` and `360` degrees (or semicolon-delimited for multi-sectors, e.g., `0; 120; 240` ).
- `Beam Tilt (°)` contains numeric tilt angles only (must match Azimuth sector count if semicolon-delimited, e.g., `2; 2; 2` ).
- Empty optional cells are left completely **BLANK** (no `"---"` , `"N/A"` , or `"null"` placeholders).

## 2. Header Structure & Data Start Row

| Row Number | Level | Description |
|---|---|---|
| **Row 1** | **Generation** | Defines cellular generations: `2G` , `3G` , `4G` , `5G` , plus antenna parameters |
| **Row 2** | **Technology** | Defines air interface standards: `GSM` , `WCDMA` , `FDD-LTE` , `TDD-LTE` , `NR` |
| **Row 3** | **Band / Column** | Core site info, specific band codes ( `B8-900` , `B3-1800` , `B28-700` , `B78-3500` ), and RF properties |
| **Row 4+** | **Data Rows** | **First actual site record begins on Row 4** |

> **Caution:** If data is mistakenly pasted starting on Row 1, 2, or 3, the entire file parser will fail immediately due to header mismatch.

## 3. Critical Active Technology Rule (Coverage Analysis)

Every cellular tower sector represents RF transceivers transmitting on specific licensed spectrum bands.

**Rule: At least ONE band column per row must be set to `TRUE` (or valid CF/BW configuration).**

| Data State in Row | System Processing Behavior | Coverage Impact |
|---|---|---|
| **At least one band is `TRUE`** (e.g., `B28-700 = TRUE`) | • Creates `Cell` (Tower site)<br>• Creates `CellSector` (Antenna structure)<br>• Creates `Cellular` (Active RF transceivers)<br>• Auto-generates `Network` & `Sector` | ✅ **Fully Functional:** Tower appears on coverage maps, wireless assessments, and connectivity analytics. |
| **ALL bands are `FALSE` or empty** | • Creates `Cell` (Tower site)<br>• Creates `CellSector` (Antenna structure)<br>❌ **Does NOT create Cellular records**<br>❌ **Does NOT create Network or Sector records** | ⚠️ **Degraded State:** Tower site is saved in inventory, but **CANNOT** be analyzed, visualized on heatmap layers, or used for signal coverage calculations. |

> **Warning:** If upload logs display: "*Pre-validation Warning: Found X records with ALL technology/band columns set to FALSE or empty*", those rows will not produce coverage analysis data. Verify the spreadsheet and set active bands to `TRUE` before uploading.

## 4. Dropdown Columns Validation

| Column Header | Allowed Values | Remarks |
|---|---|---|
| `Provider` | `DITO TELECOMMUNITY CORP.`, `GLOBE COMMUNICATIONS INC.`, `SMART COMMUNICATIONS INC.` | Select strictly from the dropdown list. |
| `Tower Type` | `GuyedTower`, `Monopole`, `Rooftop`, `SelfSupporting`, `Others`, `Unknown` | Select physical structural tower type. |
| `AntennaType` | `OmniDirectional`, `Directional`, `Others`, `Unknown` | Select antenna radiation pattern type. |
| **Band Columns** (Cols 9 to 30) | `TRUE`, `FALSE`, or blank | • Set to `TRUE` if transmitter operates on that band.<br>• Set to `FALSE` or leave blank if not equipped. |

## 5. Numeric & RF Parameters Sanitization

| Field | Example Valid Value | Invalid Entries (Will Fail) | Rule |
|---|---|---|---|
| `Lat` | `8.48222` | `8°28'56"N`, `8,48222` | Clean decimal latitude only |

| Field | Example Valid Value | Invalid Entries (Will Fail) | Rule |
|---|---|---|---|
| `Long` | `124.64722` | `124°38'50"E`, `124,64722` | Clean decimal longitude only |
| `Antenna Height (m)` | `35` or `42.5` | `35m`, `35 meters` | Numeric height in meters only |
| `Azimuth (°)` | `120` (or `0; 120; 240`) | `120 deg`, `120°`, `0, 120, 240` | Directional angle (0–360°); use semicolon `;` for multi-sectors |
| `Beam Tilt (°)` | `2` (or `2; 2; 2`) | `2 degrees`, `2 deg`, `2, 2, 2` | Mechanical/electrical tilt angle; count must match Azimuth count |

## 6. Multiple Azimuth & Beam Tilt Formatting (Multi-Sector / Tri-Sector Sites)

In cellular deployments, a physical cell tower commonly hosts multiple directional sectors (such as standard 3-sector tri-pole antennas oriented at 120° intervals, or dual-sector antennas).

The ingestion engine parses multiple sectors per site by using a **semicolon ( `;` ) delimiter**.

**Strict Ingestion Rules:**
• **Delimiter:** Separate individual sector values using a semicolon `;` (e.g., `0; 120; 240` and `2; 2; 2`). Do **not** use commas or slashes.
• **Strict 1-to-1 Count Matching:** The number of values in `Azimuth (°)` must exactly equal the number of values in `Beam Tilt (°)`. A mismatch (e.g., 3 azimuths vs 2 tilts) will abort sector creation for that row with a logged warning.
• **Pure Numbers:** Remove degree symbols ( `°` ), unit suffixes ( `deg` ), or compass directions.

| Configuration | `Azimuth (°)` | `Beam Tilt (°)` | Ingestion Behavior |
|---|---|---|---|
| **Standard Tri-Sector (3 Sectors)** | `0; 120; 240` | `2; 2; 2` | ✅ **Creates 3 CellSectors:** Sector 1: 0°/2°, Sector 2: 120°/2°, Sector 3: 240°/2° |
| **Tri-Sector (Variable Tilts)** | `30; 150; 270` | `2.5; 4.0; 2.5` | ✅ **Creates 3 CellSectors:** with their respective individual downtilts |
| **Dual Sector (2 Sectors)** | `60; 240` | `3; 3` | ✅ **Creates 2 CellSectors:** Sector 1: 60°/3°, Sector 2: 240°/3° |
| **Single Sector / Omni** | `0` | `0` | ✅ **Creates 1 CellSector:** with 0° azimuth and 0° tilt |
| ❌ **Count Mismatch (Invalid)** | `0; 120; 240` | `2; 2` | ❌ **Sector Skipped:** 3 azimuths vs 2 tilts. Sector creation skipped. |
| ❌ **Wrong Delimiter (Invalid)** | `0, 120, 240` | `2, 2, 2` | ❌ **Parse Failure:** Commas cannot be parsed as decimal lists. |

## 7. Post-Upload Support

If your file encounters processing errors:

1. Review the progress logs in the **Cell** tab of the portal.
2. Note the batch number and reported record index.

3. Check whether failure stems from missing coordinates, unrecognized provider names, count mismatch between Azimuth and Beam Tilt, or invalid non-numeric antenna heights.
4. Contact the system administrator with the file name and log excerpt for assistance.